

Act 7: The Closing Frame (Final Summary)

The Verdict on the System:

The NYC taxi ecosystem is currently in a state of **Mechanical Stability**. While there were concerns of a "quiet drift into inefficiency," our analysis of 7.54 million trips reveals a steady, cyclical rhythm that persists across both April and June.

1. Defensible Choices:

Every decision in this investigation was grounded in physical reality. We intentionally removed 9% of the raw data (752k records) because those trips represented impossible events (zero distance or time-traveling timestamps). By cleaning the "noise," we ensured our policy simulations are based on real passenger behavior.

2. Questioned Assumptions:

We initially hypothesized that tipping behavior would change based on the "human" element of the time of day (Late Night vs. Rush Hour). However, our statistical T-test (p-value 0.20) forced us to reject this. We realized our assumption was wrong: passengers aren't making emotional choices; they are following the mechanical defaults of the payment interface.

3. Limits of the Conclusion:

While the system appears stable, this analysis cannot account for external economic pressures on drivers (such as fuel costs or vehicle maintenance) or the competitive pressure from ride-sharing apps. We can say the **pulse** is steady, but we cannot say the **drivers** are thriving without further internal cost data.

Final Statement:

The signs are real, the rhythm is stable, and the system is not failing—it is simply operating as a predictable machine.